C5.3 How are the amounts of substances in reactions calculated?

Teaching and learning narrative

During reactions, atoms are rearranged but the total mass does not change. Reactions in open systems often appear to have a change in mass because substances are gained or lost, usually to the air.

Chemists use relative masses to measure the amounts of chemicals. Relative atomic masses for atoms of elements can be obtained from the Periodic Table.

The relative formula mass of a compound can be calculated using its formula and the relative atomic masses of the atoms it contains.

Counting atoms or formula units of compounds involves very large numbers, so chemists use a mole as a unit of counting. One mole contains 6.0 x 10^{23} atoms or formula units; this is the Avogadro constant. It is more convenient to count atoms as 'numbers of moles'.

In recognition of IUPAC's review, we will accept both the classical (carbon-12 based) and revised (Avogadro constant based) definitions of the mole in examinations from June 2018 onwards (see https://iupac.org/new-definition-mole-arrived/)

The number of moles of a substance can be worked out from its mass, this is useful to chemists because they can use the equations for reactions to work out the amounts of reactants to use in the correct proportions to make a particular product, or to work out which reactant is used up when a reaction stops

Assessable learning outcomes

Learners will be required to:

1. recall and use the law of conservation of mass
2. explain any observed changes in mass in non-enclosed systems during a chemical reaction and explain them using the particle model
3. calculate relative formula masses of species separately and in a balanced chemical equation
4. **recall and use the definitions of the Avogadro constant (in standard form) and of the mole**
5. **explain how the mass of a given substance (in grams) is related to the amount of that substance in moles and vice versa and use the relationship:**

   $$\textbf{number of moles} = \frac{\textbf{mass of substance}}{\textbf{relative formula mass}}$$

   M3b, M3c, M2a
6. **deduce the stoichiometry of an equation from the masses of reactants and products and explain the effect of a limiting quantity of a reactant**
7. **use a balanced equation to calculate masses of reactants or products**
   M1a, M1c

Linked learning opportunities

Specification links

- The particle model (C1.1)
- Maximising industrial yields (C6.3)

Practical Work:

- Comparison of theoretical and actual yield from the preparation of an organic compound (introduced in C3) or a salt (introduced in C5).
- Testing predictions of volumes of gases produced from reactions of acids.

Practical work:

- Making and testing predictions. Carrying out investigations. Analysing and evaluating data. Using measuring apparatus. Safe handling of chemicals.

C5.3 How are the amounts of substances in reactions calculated?

Teaching and learning narrative	Assessable learning outcomes *Learners will be required to:*		*Linked learning opportunities*
The equation for a reaction can also be used to work out how much product can be made starting from a known amount of reactants. This is useful to determine the amounts of reacting chemicals to be used in industrial processes so that processes can run as efficiently as possible. Chemists use the equation for a reaction to calculate the theoretical, expected yield of a product. This can then be compared to the actual yield. Actual yields are usually much lower than theoretical yields. This can be caused by a range of factors including reversible reactions, impurities in reactants or reactants and products being lost during the procedure. Information about actual yields is used to make improvements to procedures to maximise yields.	8.	use arithmetic computation, ratio, percentage and multistep calculations throughout quantitative chemistry M1a, M1c, M1d	**Ideas about Science:** • Using data to make quantitative predictions about yields and comparing them to actual yields. (IaS1, IaS2)
	9.	**carry out calculations with numbers written in standard form when using the Avogadro constant** M1b	
	10.	change the subject of a mathematical equation M3b, M3c	
	11.	calculate the theoretical mass of a product from a given mass of reactant *(separate science only)* M1a, M1c, M1d	
	12.	calculate the percentage yield of a reaction product from the actual yield of a reaction *(separate science only)* M1a, M1c, M1d, M3c	
	13.	suggest reasons for low yields for a given procedure *(separate science only)*	
One mole of any gas has the same volume – 24 dm^3 at room temperature and pressure. **So the number of moles of molecules in a known volume of gas can be calculated, using the formula** $\text{number of moles of gas} = \dfrac{\text{volume of gas in sample (dm}^3\text{)}}{24\,\text{dm}^3}$ **For reactions involving only gases, the relative volumes of reactant or product gases can be worked out directly from the equation.** **For reactions with substances in a mixture of states, calculations may involve using both masses and gas volumes to calculate amounts of products and reactants.**	**14.**	**describe the relationship between molar amounts of gases and their volumes and vice versa, and calculate the volumes of gases involved in reactions, using the molar gas volume at room temperature and pressure (assumed to be 24 dm^3)** ***(separate science only)***	